

Ten Edge-Degree Counts Are All a Bond-Incident-Degree Index Can See: An Exact Redundancy Certificate and What It Leaves for New Molecular Descriptors

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Abstract

On molecular graphs an edge has one of ten degree types, so every bond-incident-degree (BID) index $\sum_{uv \in E} f(d_u, d_v)$ — Zagreb, Randić, Sombor and their many variants — is a linear combination of the ten edge-type counts m_{ij} (Gutman 2013; Rada 2026). We turn this known fact into a checkable *span certificate*: a rank condition on a baseline of classical indices under which every BID index, for every f , is an exact linear combination of the baseline and can add no linear information about any target. A standard 18-index baseline satisfies it on all graphs of maximum degree four; we verify this on 13 MoleculeNet datasets (318 648 molecules), where the one failure, HIV, is caused by 350 organometallic molecules with atoms of degree up to ten. A census of 14 published degree-based indices confirms exact redundancy and exposes a numerical pitfall that turns exact zeros into spurious partial correlations. For 20 indices outside the BID class, a target-aware screen with intervals, a calibrated non-linear check and a chemistry-aware baseline finds at most two suggestive signals and none once RDKit descriptors are present. Finally, hydrogen-suppressed graphs collide (74% of FreeSolv molecules share a graph), which bounds every topology-only model below what chemistry-aware descriptors reach.

Keywords. Topological index; degree-based index; Sombor index; redundancy; partial correlation; QSPR; molecular descriptors.

Code and data. <https://github.com/Singati2/topological-index-orthogonality>

1 Introduction

Scalar topological indices have grown from a handful of early proposals [1, 2, 3] to several hundred, and the proliferation has been criticised within mathematical chemistry itself [4]. The growth is fastest for degree-based indices: since the Sombor index was introduced in 2021 [5], modified, p -, KG-, elliptic, Euler, diminished and hyperbolic Sombor variants have followed [6, 7, 8, 9, 10, 11, 12]. A new index is commonly supported by a correlation with a property on a small isomer set, by bounds and extremal graphs, and by its correlations with existing indices, e.g. intercorrelation tables among Sombor-type variants [24, 9]. Quality criteria that go beyond single correlations exist — structure sensitivity and abruptness [13], applicability tests on octanes [14], and clustering-based usefulness scores [15, 16] — but none of them answers the question a QSPR modeller needs answered: *does the new index carry information about a target that an explicit baseline does not already carry?*

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For the largest class of indices this question turns out to be settled before any data are examined. It is classical that a bond-incident-degree (BID) index $I_f(G) = \sum_{uv \in E(G)} f(d_u, d_v)$ is the linear combination $\sum_{i \leq j} f(i, j) m_{ij}(G)$ of the edge-degree-pair counts m_{ij} [4, 17], and the resulting finite-dimensional linear view of vertex-degree-based indices has recently been developed in general [18, 19], including the observation that the analytic form of f is irrelevant once its values on the degree pairs are fixed. If a baseline of classical degree-based indices spans the edge-degree-pair counts realised in a dataset — a rank condition that does not involve the candidate or the target — then *every* BID index, whatever its functional form or parameters, is an exact linear combination of the baseline on that dataset, and a target-aware redundancy screen has nothing to measure. We call this rank condition a *span certificate*: a baseline is *certified* on a dataset when the condition holds, and a candidate is then *certified redundant*.

Contributions.

- (1) *Span certificate* (Section 4). The mathematics is not ours: the linearity of BID indices in the counts m_{ij} and the finite-dimensional framework built on it are due to Gutman [4], Rada and Cruz [17] and Rada [19], and that framework contains no statistical or dataset-level analysis. What is ours is the statistical reading: an explicit, checkable rank condition on a baseline under which every BID index has zero residual, infinite variance inflation and no residual direction towards any target (Proposition 4.2); its data-free form for graphs of maximum degree at most four, with the explicit coefficient vectors (Corollary 4.3, which we have not found stated in this form); the observation that a standard 18-index baseline satisfies it; and its verification on benchmark data.
- (2) *Generality* (Section 4.1). The certificate is checked on 13 MoleculeNet SMILES datasets (318 648 molecules): it holds on 12, and the one failure (HIV) is caused by 350 organometallic molecules with atoms of degree five to ten — the case the data-free certificate excludes and the data-level check catches.
- (3) *Census* (Section 5). Fourteen published degree-based indices (15 instances, eight of them Sombor-type) and 100 points of a three-parameter family, on four MoleculeNet datasets: all exactly redundant, which sharpens earlier correlational reports of near-redundancy into an exact statement; two non-BID Sombor-type controls correctly fall outside the span.
- (4) *A numerical pitfall and a tool* (Section 6). For exactly redundant indices, partial correlations computed with absolute residual tolerances are noise of order 10^{-2} ; we give a numerically safe recipe and a one-function implementation of the certificate that returns each index’s coefficient vector.
- (5) *Target-aware screen for non-BID indices* (Section 7). Confidence intervals against an effect-size margin, a cross-fitted non-linear residualization with a proper null distribution, and a repeat of the screen against a chemistry-aware (RDKit 2D) baseline, on 20 alternative candidates.
- (6) *Information ceiling* (Section 8). Graph collisions under hydrogen suppression, confirmed by exact isomorphism, give an in-sample oracle for the whole topology-only representation; a graph-grouped, corrected benchmark places the indices against this oracle and against an atom- and bond-aware descriptor reference.

One of the parametric families screened below, the three-parameter Loyola family of Section 5, is studied mathematically in separate work by a subset of the present authors; here it serves only as a generic parametric BID family, and the span certificate explains why no such family can add linearly to a classical baseline.

2 Related work

Redundancy and orthogonalization. Strong mutual correlation among topological indices is long established. Hollas studied the redundancy of graph invariants used in chemoinformatics [20] and related the covariance of degree-based indices to the edge structure of the graph [21]; orthogonalization of descriptors is an equally old remedy [22, 23]. Recent work continues to report intercorrelations among new variants, e.g. among Sombor-type indices on octanes [9, 24], and large-scale descriptor evaluations cluster descriptors without reference to a target [15, 16]. We do not claim to discover that topological indices are redundant.

Linear structure of degree-based indices. The identity $I_f = \sum f(i, j)m_{ij}$ is standard [4, 17] and underlies the bond-additive modelling programme in which the term “bond incident degree” index originates [25, 26]. Matrix and finite-dimensional linear formulations of vertex-degree-based indices, including their discrimination properties, are developed in [18, 19]; the M-polynomial [27] is the generating function of the same counts; further algebraic constructions on degree-based indices appear in [28]; and the edge-type-count view underlies recent extremal-graph work for chemical graphs of maximum degree three [31, 29, 30]. The boundary between that literature and the present paper is stated once, in Contribution (1).

Degeneracy. That distinct graphs can share index values (degeneracy) and that hydrogen suppression merges distinct molecules are classical concerns [33, 34, 35]; a recent audit of the MoleculeNet benchmarks documents duplicate and conflicting entries at the SMILES level [32]. Section 8 quantifies collisions at the level of the hydrogen-suppressed graph (isomorphism classes), which is the relevant level for topological indices, and uses them as a ceiling for the whole topology-only representation rather than for a single index.

Descriptor evaluation in QSPR. Topological descriptors remain competitive baselines in modern property prediction [36, 37], which keeps the question of which indices are worth adding practically relevant. Generic feature selection (LASSO [38], mRMR [39], Boruta [40]) acts on a fixed descriptor matrix; the screen here acts one layer earlier, on whether a proposed index should enter that matrix at all.

3 Data, curation and baseline

We use four MoleculeNet benchmarks [41]: ESOL (aqueous solubility) [42], FreeSolv (experimental hydration free energy; column `expt`) [43], Lipophilicity ($\log D_{7.4}$) [44] and BBBP (blood–brain barrier penetration, binary) [45]. SMILES are parsed with RDKit [46]; the largest fragment is kept, the hydrogen-suppressed, atom- and bond-unlabelled graph is formed, and molecules with fewer than two heavy atoms are dropped, so that every graph is connected and has no isolated vertex. On the files as distributed this leaves 1127, 639, 4200 and 2039 molecules (the “parsed” column of Table 1). Table 1 reports the counts. Stereochemistry and atom identity are invisible to every descriptor studied here, so molecules that differ only in these respects become duplicates; we report all screening results on the files as distributed and verify in Section 7 that de-duplication on canonical SMILES (averaging regression duplicates whose labels agree within 0.5 target units, dropping duplicate groups that disagree by more or carry conflicting BBBP labels) changes no conclusion; the benchmark of Section 8 uses the curated data. All molecules have maximum degree at most four after hydrogen suppression.

Baseline. The 30-index baseline (Supplementary Table S1) contains 18 degree-based indices, 8 distance-based indices, 3 spectral indices and the Balaban index. All 18 degree-based indices are BID indices in the sense of Section 4 (vertex-degree sums $\sum_v g(d_v)$ are BID with $f(a, b) =$

Table 1: Datasets and curation. Raw rows of the MoleculeNet files; invalid SMILES; multi-fragment SMILES reduced to the largest fragment; molecules parsed to a graph with at least two heavy atoms (the sets used by the screens); rows in exact canonical-SMILES duplicate groups; duplicate groups dropped for conflicting labels; molecules after curation (also dropping graphs with fewer than two heavy atoms); distinct hydrogen-suppressed unlabelled graphs (Weisfeiler–Lehman hash); and the percentage of curated molecules whose graph is shared with another molecule.

Dataset	Target	raw	invalid	multi-frag.	parsed	dup. rows	conflicts	curated	graphs	shared (%)
ESOL	$\log S$ (mol/L)	1128	0	0	1127	22	1	1115	628	55.5
FreeSolv	ΔG_{hyd} (kcal/mol)	642	0	0	639	0	0	639	236	74.3
Lipophilicity	$\log D_{7.4}$	4200	0	1	4200	0	0	4200	3472	28.2
BBBP	BBB penetration (0/1)	2050	11	105	2039	141	11	1955	1693	22.6

$g(a)/a + g(b)/b$ on graphs without isolated vertices). Several baseline members are related by exact identities, which we disclose because they matter for numerical rank: $\sigma = F - 2M_2$ and $\text{red}M_1 = M_1 - 4|E(G)| + |V(G)|$. The index labelled ‘‘Schultz’’ in our code is the degree distance $\sum_{\{u,v\}} (d_u + d_v) d(u,v)$ (the Schultz molecular topological index differs by M_1); PI is the edge Padmakar–Ivan index; the augmented Zagreb summand is set to zero on K_2 edges.

The composition of the baseline is listed in Supplementary Table S1.

4 The span certificate

Let $\mathcal{G}_\Delta$ be the class of finite simple graphs without isolated vertices and with maximum degree at most Δ , and $P_\Delta = \{(i,j) : 1 \leq i \leq j \leq \Delta\}$, so $|P_\Delta| = 10$. For $G \in \mathcal{G}_\Delta$ let $m_{ij}(G)$ be the number of edges whose end degrees are $\{i,j\}$. A *BID index* is $I_f(G) = \sum_{uv \in E(G)} f(d_u, d_v)$ for a symmetric $f : \{1, \dots, \Delta\}^2 \rightarrow \mathbb{R}$.

Remark 4.1 (classical). $I_f(G) = \sum_{(i,j) \in P_\Delta} f(i,j) m_{ij}(G)$, because the edge classes $\{uv : \{d_u, d_v\} = \{i,j\}\}$ partition $E(G)$ and f is constant on each [4, 17]. For example, the path on four vertices has two edges of type $(1,2)$ and one of type $(2,2)$, so $I_f = 2f(1,2) + f(2,2)$ on it for every f , and the star $K_{1,3}$ has three edges of type $(1,3)$, so $I_f(K_{1,3}) = 3f(1,3)$. On $\mathcal{G}_4$ the ten functionals m_{ij} are linearly independent: the complete bipartite graph $K_{i,j}$ has all ij edges of type (i,j) , so its count vector is a multiple of the (i,j) -th unit vector. Hence the BID indices on $\mathcal{G}_4$ form exactly a ten-dimensional space; on connected graphs with at least three vertices $m_{11} = 0$ and the space is at most nine-dimensional.

Now fix a dataset $D = \{G_1, \dots, G_n\} \subset \mathcal{G}_\Delta$ with realised pair set $P_D \subseteq P_\Delta$, count matrix $M \in \mathbb{R}^{n \times |P_D|}$, and a baseline whose BID block consists of k indices with value matrix $\Phi_B \in \mathbb{R}^{k \times |P_D|}$, $(\Phi_B)_{b,(i,j)} = f_b(i,j)$. The baseline’s BID columns are $B = M\Phi_B^\top \in \mathbb{R}^{n \times k}$. Write $\mathbf{1}$ for the all-ones vector.

The following is an elementary consequence of Remark 4.1; its statistical reading (i)–(iv) is the point.

Proposition 4.2 (Span certificate). *If $\text{rank}[\mathbf{1}, B] = \text{rank}[\mathbf{1}, M]$, then for every symmetric f the vector $z_f = (I_f(G_t))_{t \leq n}$ lies in $\text{col}[\mathbf{1}, B]$. Consequently, for any baseline matrix X whose columns include B and for every target vector y : (i) the least-squares residual of z_f on $[\mathbf{1}, X]$ is zero; (ii) if z_f is not constant on D , its variance inflation relative to X is infinite; (iii) the partial correlation $\text{pcor}(z_f, y \mid X)$ — the correlation between the residuals of z_f and of y after least-squares regression on $[\mathbf{1}, X]$ — is undefined (0/0): z_f carries no direction beyond X , and we report it as 0 by convention — any nonzero value computed in floating point is an artefact (Section 6); and (iv) adding z_f to X leaves every ordinary least-squares fit (fitted*

values, residuals, R^2) on D unchanged. Penalised linear fits (ridge, LASSO) can change, but only through the geometry of the penalty, not through any new direction in the data.

Proof. By Remark 4.1, $z_f = Mf_D$ with $f_D = (f(i, j))_{(i, j) \in P_D}$, so $z_f \in \text{col}(M)$. Since $B = M\Phi_B^\top$, $\text{col}[\mathbf{1}, B] \subseteq \text{col}[\mathbf{1}, M]$, and equal ranks give equality. Thus $z_f \in \text{col}[\mathbf{1}, B] \subseteq \text{col}[\mathbf{1}, X]$, which is (i); (ii) and (iii) restate (i), and (iv) holds because an OLS fit depends on the design only through its column space. $\square$

Corollary 4.3 (data-free certificate on $\mathcal{G}_4$). *If $\text{rank } \Phi_B = 10$ over P_4 , then for every symmetric f there is a coefficient vector c , unique up to $\ker \Phi_B^\top$, with $\Phi_B^\top c = f_{P_4}$, and $I_f(G) = \sum_b c_b I_{f_b}(G)$ holds identically on $\mathcal{G}_4$. In particular the conclusions of Proposition 4.2 hold on every dataset of graphs in $\mathcal{G}_4$, without inspecting the data.*

Proof. $\text{rank } \Phi_B = 10$ means the rows f_b span $\mathbb{R}^{P_4}$; then

$$I_f = \sum_{ij} f(i, j) m_{ij} = \sum_{ij} \sum_b c_b f_b(i, j) m_{ij} = \sum_b c_b I_{f_b}.$$

$\square$

The baseline satisfies the certificate. Evaluating each of the 18 degree-based baseline indices on $K_{i,j}$ gives Φ_B exactly (Supplementary Table S2); its rank over P_4 is 10 (smallest-to-largest singular value ratio 1.2×10^{-4}), so Corollary 4.3 applies. The certificate is generic rather than delicate: 37 322 of the 43 758 ten-element subsets of the 18 indices already have rank 10, for example $\{M_1, M_2, mM_1, mM_2, F, R, \text{SCI}, H, GA, AG\}$. On the data, the numerical ranks of M , B and $[B, M]$ coincide on all four datasets (Table 2).

Remark 4.4 (what the certificate does and does not say). The certificate is a statement about *linear* redundancy. It does not say that a BID index is useless: a parsimonious index can summarise the ten counts well, and a model that is not rotation-invariant (e.g. a tree ensemble) may use a BID index differently from its constituents. It does say that no BID index can add linearly to a certified baseline, so linear redundancy diagnostics — pairwise correlation, the variance inflation factor (VIF), partial correlation, and the gain of an ordinary least-squares model — are decided in advance for this class. The certificate covers edge-additive indices only; it does not cover normalised or averaged indices such as $I_f/|E|$, multiplicative indices $\prod f$, non-linear transforms such as $\log I_f$, products of indices, neighbourhood-degree-sum and other second-order indices, or valence- and heteroatom-weighted degrees. Beyond $\Delta = 4$ (hypervalent atoms, explicit hydrogens, coordination compounds) the space has dimension $\Delta(\Delta + 1)/2$ and the certificate must be checked on the data, which Proposition 4.2 makes a one-line rank computation.

Table 2: Span certificate on the four datasets. $|P_D|$: realised degree pairs; ranks are numerical ranks of the column-centred, column-normalised matrices (relative tolerance 10^{-9}). “Resid.” columns give the largest relative residual $\|r\|/\|z - \bar{z}\|$ on the baseline of the ten counts m_{ij} , of the 15 published BID instances of Table 4, and of the 100 Loyola grid points.

Dataset	n	$ P_D $	rk M	rk B	rk $[B, M]$	resid. m_{ij}	resid. published	resid. Loyola
ESOL	1127	10	10	10	10	4.5×10^{-13}	1.9×10^{-13}	6.9×10^{-13}
FreeSolv	639	10	10	10	10	2.1×10^{-12}	5.6×10^{-14}	1.3×10^{-13}
Lipophilicity	4200	9	9	9	9	2.6×10^{-12}	3.5×10^{-13}	1.9×10^{-12}
BBBP	2039	10	10	10	10	1.1×10^{-12}	1.8×10^{-13}	1.3×10^{-12}

4.1 Scope check across MoleculeNet

The data-free certificate covers every dataset of graphs with maximum degree four. To see how far that reaches, and what happens beyond it, we ran the data-level check of Proposition 4.2 on every SMILES-based MoleculeNet dataset we could obtain (Table 3; only SMILES are used, no targets; 318 648 molecules in all). For the eight datasets whose molecules all have maximum degree four — including the 133 882 molecules of QM9 and the 93 087 of MUV — Corollary 4.3 guarantees the outcome, and the run is a numerical check of it: the ranks of M and B agree and every m_{ij} column has relative residual at most 1.6×10^{-9} . The informative cases are the other five. Four datasets (ClinTox, SIDER, Tox21, ToxCast) contain one to three molecules with an atom of degree five or six; here certification is data-dependent, because as functions on P_6 the 18 baseline functions have rank only 17 ($\sigma = F - 2M_2$ exactly), so at most seven pairs beyond P_4 can ever be absorbed. These datasets realise one to three such pairs, the baseline’s rank rises by the same amount, and the certificate holds. The single failure is HIV: 350 of its 41 127 molecules are organometallic complexes (Co, Ni, Cu, Rh, Fe, Mn, Cr, Mo, Ir centres with five to ten bonds, and pentavalent phosphorus), which realise 37 degree pairs; the count matrix has rank 36 while the 18-index baseline has rank 17, so no baseline of 18 BID indices can certify this dataset. On the 40 777 HIV molecules of maximum degree four the certificate holds again (ranks 10/10, residual 5.0×10^{-12}). This is the intended behaviour: the data-free certificate is a statement about $\mathcal{G}_4$, and the one-line data-level rank check is what detects a dataset that leaves that class.

Table 3: The span certificate on thirteen MoleculeNet SMILES datasets (no targets used). n : parsed molecules; $n_{>4}$: molecules containing an atom of degree above four; Δ : largest degree; $|P_D|$: realised degree pairs; ranks of the count matrix M and of the 18-index baseline block B ; largest relative residual of an m_{ij} column on B ; whether $\text{rank}[\mathbf{1}, B] = \text{rank}[\mathbf{1}, M]$ with residuals below 10^{-8} .

Dataset	n	$n_{>4}$	Δ	$ P_D $	rk M	rk B	max resid.	certified
ESOL	1127	0	4	10	10	10	4.6×10^{-13}	yes
FreeSolv	639	0	4	10	10	10	2.1×10^{-12}	yes
Lipophilicity	4200	0	4	9	9	9	2.6×10^{-12}	yes
BBBP	2039	0	4	10	10	10	1.1×10^{-12}	yes
BACE	1513	0	4	9	9	9	1.8×10^{-12}	yes
ClinTox	1477	1	6	11	11	11	2.4×10^{-12}	yes (on the $\Delta \leq 4$ subset, $n = 1476$: ranks 10/10, yes)
SIDER	1396	1	6	11	11	11	7.7×10^{-12}	yes (on the $\Delta \leq 4$ subset, $n = 1395$: ranks 10/10, yes)
Tox21	7812	2	6	12	12	12	6.6×10^{-11}	yes (on the $\Delta \leq 4$ subset, $n = 7810$: ranks 10/10, yes)
ToxCast	8566	3	6	13	13	13	1.6×10^{-9}	yes (on the $\Delta \leq 4$ subset, $n = 8563$: ranks 10/10, yes)
QM8	21783	0	4	10	10	10	3.6×10^{-11}	yes
QM9	133882	0	4	10	10	10	1.4×10^{-10}	yes
MUV	93087	0	4	9	9	9	2.0×10^{-11}	yes
HIV	41127	350	10	37	36	17	9.9×10^{-1}	no (on the $\Delta \leq 4$ subset, $n = 40777$: ranks 10/10, yes)

A one-function implementation. The certificate and the coefficient vectors are available as a single function, `certify(molecules, f, baseline)` in `src/certify.py`, which returns the data-free rank of Φ_B , the realised pairs, the data-level ranks and residuals, and, for a candidate f , its coefficient vector on the baseline with the identity error on the data. For the Euler Sombor index on five test molecules the identity error is 7×10^{-16} ; for the hyper-Zagreb index on the ten-index basis it returns $HM = F + 2M_2$ exactly. The 18×10 value matrix Φ_B is listed in the Supplementary Information (Table S2).

5 Census of published degree-based indices

We screened the 15 instances of 14 published BID indices in Table 4 — eight Sombor-type instances (the p -Sombor index at two values of p) and seven other degree-based indices — and all 100 points of the Loyola grid $(\alpha, \beta, \gamma) \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}^2 \times \{0, \frac{1}{2}, 1, 2\}$ of $LO(G; \alpha, \beta, \gamma) = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp(\gamma |d_u - d_v| / (d_u + d_v))$ (a three-parameter BID family that reduces to the Gutman–Milovanović family at $\gamma = 0$) on each of the four datasets. As controls outside the BID class we added the Sombor coindex (sum over non-adjacent pairs) and the eccentric Sombor index (eccentricities in place of degrees).

Table 4: Census of published BID indices ($f(a, b)$ with a, b the end degrees of an edge) and two non-BID Sombor-type controls. Columns 3–6: largest $|r|$ with any of the 30 baseline indices on each dataset (the conventional pairwise screen). Last column: largest relative residual on the baseline over the four datasets. For KG-Sombor $e = a + b - 2$ is the edge degree. Every BID index is in the span to machine precision; the controls are not.

Index	$f(a, b)$	ESOL	FreeSolv	Lipo	BBBP	max resid.
Modified Sombor [6]	$1/\sqrt{a^2 + b^2}$	0.999	0.999	0.999	1.000	8.6×10^{-14}
Elliptic Sombor [7]	$(a + b)\sqrt{a^2 + b^2}$	1.000	0.999	0.999	1.000	4.1×10^{-14}
Euler Sombor [8]	$\sqrt{a^2 + b^2 + ab}$	1.000	1.000	1.000	1.000	6.8×10^{-14}
Diminished Sombor [9]	$\sqrt{a^2 + b^2}/(a + b)$	1.000	1.000	1.000	1.000	4.6×10^{-14}
Hyperbolic Sombor [10]	$\sqrt{a^2 + b^2}/\min(a, b)$	0.987	0.983	0.985	0.990	3.5×10^{-13}
p -Sombor, $p = \frac{1}{2}$ [11]	$(a^{1/2} + b^{1/2})^2$	1.000	1.000	1.000	1.000	5.9×10^{-14}
p -Sombor, $p = 3$ [11]	$(a^3 + b^3)^{1/3}$	1.000	1.000	1.000	1.000	4.6×10^{-14}
KG-Sombor [12]	$\sqrt{a^2 + e^2} + \sqrt{b^2 + e^2}$	0.999	0.999	0.999	0.999	1.1×10^{-13}
Inverse sum indeg [25]	$ab/(a + b)$	0.998	0.997	0.998	0.998	1.2×10^{-13}
Symmetric division deg [25]	$a/b + b/a$	0.993	0.992	0.992	0.994	1.1×10^{-13}
Nirmala [54]	$\sqrt{a + b}$	1.000	1.000	1.000	1.000	4.7×10^{-14}
First inverse Nirmala [55]	$\sqrt{1/a + 1/b}$	0.999	0.999	0.999	0.999	4.9×10^{-14}
Second inverse Nirmala [55]	$\sqrt{ab/(a + b)}$	1.000	0.999	0.999	1.000	1.8×10^{-13}
First Gourava [56]	$a + b + ab$	0.999	0.999	0.999	0.999	4.6×10^{-14}
Hyper-Zagreb [57]	$(a + b)^2$	0.999	0.998	0.998	0.999	4.1×10^{-14}
Sombor coindex (control)	non-BID	0.999	0.998	0.997	0.999	7.7×10^{-3}
Eccentric Sombor (control)	non-BID	0.971	0.976	0.943	0.947	7.5×10^{-2}

Every published BID index lies in the span of the baseline to a relative residual of at most 4×10^{-13} , and every Loyola grid point to at most 2×10^{-12} ; the two controls have relative residuals of 0.006–0.008 (coindex) and 0.06–0.08 (eccentric Sombor), i.e. they are correctly *not* certified. The universal coefficients of Corollary 4.3 reproduce the data-level values of all 15 instances to relative error below 10^{-10} (file `results/bid_universal_identities.csv`). Two of them reduce to familiar identities, $GG_1 = M_1 + M_2$ and $HM = F + 2M_2$; the others have non-sparse coefficient vectors, which is the generic situation and the reason the redundancy is invisible to a pairwise inspection.

The census is a numerical verification of Corollary 4.3, not independent evidence for it. The pairwise screen happens to reject all 15 published instances here (their maximal correlation with a baseline index is 0.98–1.00), and it passes only one of the 400 Loyola evaluations (100 grid points on four datasets) (FreeSolv, $(\alpha, \beta, \gamma) = (-1, -1, 1)$, $\max |r| = 0.943$). The certificate reaches the same verdicts for a different reason: it is exact, needs no threshold, and would apply equally to a variant whose pairwise correlations happened to fall below any threshold.

6 A numerical pitfall

An earlier draft of this work reported maximal partial correlations of 0.028 (ESOL), 0.036 (FreeSolv) and 0.009 (Lipophilicity) for the Loyola grid. By Proposition 4.2 the true value is

zero. The nonzero values arose because a residual was declared zero only when its absolute standard deviation fell below 10^{-12} ; for indices of magnitude 10^2 – 10^4 the least-squares residual of an exactly redundant index is floating-point noise of relative size 10^{-15} – 10^{-12} but absolute size above that tolerance, and correlating this noise with the target residual gives values of order $1/\sqrt{n}$. Partial correlations are the most affected, because they are a ratio of two noise quantities; VIF and incremental R^2 computed in the same way appear as finite but meaningless numbers (very large, respectively very small). We recommend (a) a relative criterion $\|r_z\|/\|z - \bar{z}\| < 10^{-8}$ with a rank-revealing solver, (b) reporting the in-span status explicitly, and (c) for BID candidates, replacing the statistical computation by the certificate. Our implementation (`src/orthogonality.py`, `span_certified_pcor`) does all three; with it, all 400 Loyola evaluations are exactly 0.

7 A target-aware screen for indices outside the BID class

For a candidate z that is not certified, the question is whether it carries information about a target y beyond the baseline X . We report three quantities.

- (a) *Pairwise screen*: $\max_b |r(z, X_b)| < 0.95$, kept only for comparability with current practice.
- (b) *Linear partial correlation with a confidence interval*: $\text{pcor}(z, y \mid X)$ from OLS residuals on $[\mathbf{1}, X]$, with a Fisher- z interval with standard error $1/\sqrt{n - \text{rk } X} - 3$, where $\text{rk } X$ is the numerical rank of X , read against an effect-size margin $\tau = 0.10$ fixed before the analysis: *pass* if the 95% interval for $|\text{pcor}|$ lies above τ , *negligible* if it lies below, *inconclusive* otherwise. The margin is a convention for a “one percent of residual variance” effect, not a significance threshold.
- (c) *Non-linear conditional association*: z and y are each residualized on X by cross-fitted random forests (five folds), and the correlation ρ_{nl} of the two residual vectors is reported with its Fisher- z interval, in the spirit of double machine learning and the generalised covariance measure [49, 50]. Because the forests are only approximate, we calibrate against three references computed in the same way: an exactly redundant BID index, a pure-noise column, and a deliberately leaky column $y + \varepsilon$.

The 20 candidates span information-theoretic, centrality, spectral-hybrid, motif and eccentricity families (Supplementary Information, Section S3); none is claimed as new, and none is a BID index. (Harmonic centrality is identically twice the Harary index and is correctly flagged as in span.)

Results. The pairwise screen admits 15, 15, 16 and 17 of the 20 candidates on ESOL, FreeSolv, Lipophilicity and BBBP. Point estimates $|\text{pcor}| \geq 0.10$ would add four ESOL candidates and one FreeSolv candidate, but with intervals none of them clears the margin: they are inconclusive, with intervals straddling 0.10 (Table 5). This is largely a matter of power: the half-width of a 95% interval is about 0.08 at $n = 639$ (FreeSolv) and 0.06 at $n = 1127$ (ESOL), so a pass would require $|\text{pcor}|$ of about 0.18 and 0.16. On Lipophilicity ($n = 4200$) 18 of the 20 are negligible. Only on BBBP do two candidates, the triangle and four-cycle counts, clear the margin with their whole interval (linear-probability partial correlations 0.225 and -0.220).

The non-linear check separates these further. The calibration references behave as intended: the leaky column gives $\rho_{\text{nl}} = 0.57$ – 0.66 on every dataset, while the exactly redundant reference and the noise column stay small. Because the random-forest residualization is not perfect — on Lipophilicity even the two null references have intervals that exclude zero — we drew a null distribution for ρ_{nl} with the same procedure: 500 random BID indices Mc , $c \sim N(0, I)$, which are exactly in the linear span of the baseline (the relevant null for a redundant index), and 100 pure-noise columns, per dataset. The 99th percentile of $|\rho_{\text{nl}}|$ under the in-span null is 0.11 (ESOL), 0.11 (FreeSolv), 0.08 (Lipophilicity) and 0.14 (BBBP); under pure noise it

Table 5: Target-aware screen of the 20 non-BID candidates. Top: counts per dataset — pairwise pass ($\max |r| < 0.95$), point-estimate combined pass ($|\text{pcor}| \geq 0.10$ and pairwise pass), interval pass (pairwise pass and 95% interval of $|\text{pcor}|$ above 0.10), inconclusive, negligible, and in span. Bottom: every candidate–dataset pair with $|\text{pcor}| \geq 0.10$ or $|\rho_{\text{nl}}| \geq 0.10$, with 95% Fisher- z intervals.

Dataset	pairwise	point comb.	interval pass	inconcl.	neglig.	in span
ESOL	15	4	0	12	7	1
FreeSolv	15	1	0	15	4	1
Lipophilicity	16	0	0	1	18	1
BBBP	17	2	2	1	16	1

Dataset	Candidate	$\max r $	linear pcor [95% CI]	verdict	ρ_{nl} [95% CI]
ESOL	Bonchev_Id	0.992	−0.113 [−0.171, −0.054]	inconclusive	+0.015 [−0.043, +0.074]
ESOL	CutVerts	0.833	−0.118 [−0.176, −0.059]	inconclusive	−0.015 [−0.074, +0.043]
ESOL	FourCyc	0.189	+0.112 [+0.054, +0.170]	inconclusive	+0.001 [−0.058, +0.059]
ESOL	InfoH_deg	0.699	+0.154 [+0.096, +0.211]	inconclusive	+0.142 [+0.085, +0.199]
ESOL	MeanEcc	0.930	−0.109 [−0.167, −0.050]	inconclusive	−0.028 [−0.087, +0.030]
FreeSolv	EE_x_W	0.985	+0.129 [+0.051, +0.206]	inconclusive	+0.027 [−0.050, +0.105]
FreeSolv	InfoH_deg	0.649	−0.061 [−0.140, +0.018]	inconclusive	−0.104 [−0.181, −0.027]
FreeSolv	Triangles	0.126	+0.107 [+0.028, +0.184]	inconclusive	+0.011 [−0.067, +0.089]
BBBP	FourCyc	0.224	−0.220 [−0.261, −0.178]	pass	−0.171 [−0.212, −0.128]
BBBP	Triangles	0.119	+0.225 [+0.183, +0.266]	pass	+0.013 [−0.031, +0.056]

Calibration of ρ_{nl} across the four datasets: exactly redundant BID reference ($\sum (d_u + d_v)^{-1/2}$, i.e. SCI) −0.044 to +0.043; pure noise −0.030 to +0.039; leaky column $y + \varepsilon$ 0.567 to 0.661.

is 0.08, 0.10, 0.05 and 0.05. The null width is therefore sampling variance ($\approx 2.6/\sqrt{n}$) plus, on ESOL, Lipophilicity and BBBP, an additional contribution from columns that lie in the span but that a forest residualizes imperfectly; on FreeSolv the two nulls coincide. Any residual association must be judged against this null width, not against zero (Figure 1b). Two candidate–dataset pairs exceed every one of the 500 in-span draws on their dataset: the degree entropy InfoH_deg on ESOL ($\rho_{\text{nl}} = 0.14$, [0.08, 0.20]; $p = 0.002$) and the four-cycle count on BBBP (−0.17, [−0.21, −0.13]; $p = 0.002$); a third, InfoH_deg on FreeSolv (−0.10, [−0.18, −0.03]), is exceeded by nine draws ($p = 0.018$). After Benjamini–Hochberg adjustment over the 80 pairs the first two reach $q = 0.08$ and the third $q = 0.48$; 0 pairs are significant at $q < 0.05$. The two strongest also have linear intervals that exclude zero, so they are the only pairs with support from both residualizations; we call them suggestive. (The empirical p -values use the 100-tree statistic of the null run; Table 5 reports the 300-tree values, which differ by at most 0.004.) Several linear “signals” vanish once the baseline is allowed to act non-linearly — most strikingly the BBBP triangle count (0.225 $\rightarrow$ 0.013) — which means that their linear residual association was information the baseline already carries in non-linear form. Permutation p -values of a distance correlation between the residuals were not calibrated (the exactly redundant reference was rejected on two datasets) and are not used.

Against a chemistry-aware baseline. A QSPR practitioner’s baseline is not thirty topological indices but a standard descriptor set. We therefore repeated the linear screen with the baseline extended by RDKit 2D descriptors (the 180 of Section 8, of which 149–162 remain after removing non-finite, constant and collinear columns; none of the 30 topological indices is collinear with them) (Table 6). Against the union baseline 0 of the 20 candidates passes on any dataset. The two candidates with the largest statistics above become negligible; on Lipophilicity and BBBP most candidates are negligible, whereas on ESOL and FreeSolv most remain unresolved at these sample sizes (interval half-widths of about 0.06 and 0.09 with a baseline of

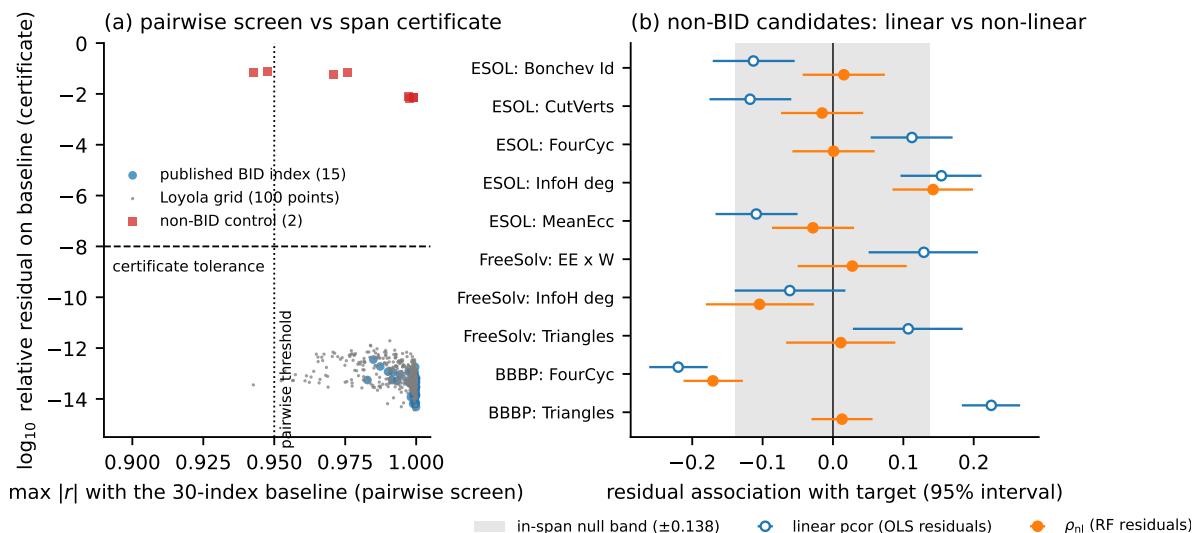


Figure 1: (a) The conventional pairwise screen (largest $|r|$ with any of the 30 baseline indices) cannot separate the 15 published BID instances and the 100 Loyola points (all exactly in span, residual 10^{-15} – 10^{-12}) from the two non-BID controls (residual 10^{-2} – 10^{-1}): all lie at $|r| \geq 0.94$. The certificate separates them by ten orders of magnitude. (b) The candidate–dataset pairs of Table 5: linear partial correlation (hollow) and non-linear ρ_{nl} (filled) with 95% intervals; the shaded band is the widest of the four in-span null widths (99th percentile of $|\rho_{nl}|$ over 500 exactly redundant draws; per-dataset values in the text). ESOL/InfoH_deg and BBBP/FourCyc exceed every draw on their own dataset ($p = 0.002$, $q = 0.08$ over the 80 pairs); no pair reaches $q < 0.05$.

rank about 180), so “none passes” is the accurate summary, not “all are negligible”. The reverse question has almost the same answer: of the 30 topological indices, 1 (the Mostar index on ESOL) retains an interval for $|\text{pcor}|$ above 0.10 given the RDKit block, whereas 18–72 RDKit descriptors per dataset retain such signal given the topological block. Out of sample, adding all 30 indices and all 20 candidates to the RDKit descriptors improves a ridge model by at most 0.024 RMSE (ESOL; 3% of the reference, a point estimate from a single 5-fold split without an interval). Topology-only indices, new or old, add little to a chemistry-aware baseline on these targets.

Table 6: The target-aware screen against a chemistry-aware baseline. p_{RD} : RDKit 2D descriptors retained (finite, non-constant, not collinear); p_{both} : columns of the union with the 30 topological indices after removing exact collinearity (none of the 30 was collinear with the RDKit block). R^2 : in-sample OLS. CV: 5-fold cross-validated RMSE (ridge, penalty by inner CV) or ROC-AUC (BBBP, ℓ_2 -logistic). Candidates: interval verdicts given the union baseline (pass / inconclusive / negligible; the remaining candidate is in span). Last column: number of the 30 topological indices whose interval for $|\text{pcor}|$ given the RDKit block lies above 0.10.

Dataset	p_{RD}	p_{both}	in-sample R^2			CV RMSE / AUC			candidates pass/inc./negl.	topo. beyond RD
			RD	RD+topo	RD+topo+cand	RD	RD+topo	RD+topo+cand		
ESOL	161	191	0.929	0.940	0.944	0.729	0.731	0.705	0 / 13 / 6	1
FreeSolv	149	179	0.972	0.976	0.977	1.051	1.044	1.043	0 / 18 / 1	0
Lipophilicity	158	188	0.591	0.607	0.617	0.829	0.824	0.820	0 / 3 / 16	0
BBBP	162	192	0.545	0.559	0.566	0.891	0.892	0.894	0 / 1 / 18	0

De-duplication. Repeating the linear screen on the curated datasets of Table 1 changes $|\text{pcor}|$ by at most 0.009 and changes no verdict.

8 The information ceiling of topology-only descriptors

Every descriptor considered in this paper — all 30 baseline indices, all BID indices, and all 20 alternative candidates — is a function of the hydrogen-suppressed, atom- and bond-unlabelled molecular graph. Molecules with the same graph therefore receive identical descriptor vectors, whatever their atoms, bonds or stereochemistry. We grouped the curated molecules by a Weisfeiler–Lehman hash [51] of this graph (a canonical label built by four rounds of neighbourhood-label aggregation, identical for isomorphic graphs), and confirmed with an exact isomorphism test that every one of the 841 groups of molecules sharing a hash consists of isomorphic graphs; the grouping is therefore exact.

The collisions are extensive (Table 1): 55.5% of the curated ESOL molecules and 74.3% of the FreeSolv molecules share their graph with at least one other molecule; FreeSolv’s 639 molecules have only 236 distinct graphs, and the largest group contains 24 molecules. Within groups of molecules that share a graph, the pooled standard deviation of the target is 1.29 (ESOL), 3.18 (FreeSolv) and 0.62 (Lipophilicity), against overall standard deviations of 2.09, 3.85 and 1.20; on BBBP, 15.6% of the shared groups contain both labels. Because no topology-only model can separate molecules within a group, this within-group variance cannot be reduced by any such model (it also contains experimental noise, which limits every model). Note that the pooled within-group standard deviation is not the oracle’s error: molecules with a unique graph contribute zero within-group variance, which is why the oracle RMSE below is smaller. The sharpest statement uses the *graph oracle*: the in-sample predictor that assigns every molecule the mean target of its isomorphism class. No function of the unlabelled graph can beat it on the same data. Its R^2 is 0.83 (ESOL), 0.57 (FreeSolv) and 0.95 (Lipophilicity), with RMSE 0.85, 2.53 and 0.26 (Figure 2); it is an in-sample bound for a dataset, not a population quantity, and it is loose where most molecules have a unique graph (Lipophilicity: 28% shared). The out-of-sample counterpart — predicting each molecule from the other members of its class, and by the global mean when it has none — reaches only RMSE 1.87, 3.87 and 1.11, which shows that most of the information in these benchmarks is *not* carried by graph identity alone. The oracle nevertheless settles the FreeSolv question: the best topology-only model (Table 7) explains far less than the oracle allows, and the chemistry-aware reference below explains more out of sample ($R^2 \approx 0.89$, computed as $1 - \text{RMSE}^2 / \text{Var}(y)$ from Table 7) than any topology-only model could in sample (0.57); the same holds on ESOL (0.90 versus 0.83). The comparison is fair in the direction that matters: with folds grouped by graph, the cross-validated predictions of any topology-only model are constant within an isomorphism class, so its cross-validated error is at least the in-sample oracle’s. The weak performance of topological indices on FreeSolv is therefore a property of the representation, not of any particular index.

Benchmark. To place the indices against this ceiling we ran a downstream benchmark on the curated data (Table 7). Folds are grouped by graph so that identical descriptor vectors never appear on both sides of a split; the design is 5×5 repeated cross-validation; standardisation, screening, and the choice of the LASSO penalty all happen inside the training folds; and paired differences are tested with the Nadeau–Bengio variance correction [52], with equivalence assessed by two one-sided tests (TOST) [53] against a margin of 5% of the reference RMSE, or $0.05(\text{AUC}_{\text{ref}} - 0.5)$ for BBBP; the two one-sided tests at level 0.05 correspond to the reported 90% intervals. As a chemistry-aware reference we used a random forest on RDKit two-dimensional descriptors (180 of RDKit’s 208).¹

¹The remaining 28 (EState- and VSA-EState-type descriptors, BalabanJ, BertzCT and Ipc) terminate the process under the RDKit 2022.09 / NumPy 2 combination used here and were excluded. Three of them are

Table 7: Downstream benchmark on the curated data: 5×5 repeated cross-validation with folds grouped by graph, all preprocessing and selection inside the training folds. Top: mean RMSE (ESOL, FreeSolv, Lipophilicity) or ROC-AUC (BBBP), with the mean number of active features in parentheses. Bottom: mean paired difference (model minus reference; lower RMSE and higher AUC are better) with Nadeau–Bengio corrected 90% intervals; “eq.” marks equivalence (TOST) within 5% of the reference RMSE or 5% of the reference AUC above 0.5. *On Lipophilicity the combined screen kept no feature in 24 of 25 folds (intercept-only model).

Model	ESOL	FreeSolv	Lipophilicity	BBBP (AUC)
full 30, LASSO / ℓ_1 -logistic	1.535 (16.0)	3.516 (11.6)	1.130 (13.9)	0.808 (27.1)
pairwise-screened, LASSO	1.580 (7.2)	3.515 (5.9)	1.139 (6.9)	0.791 (5.8)
combined-screened, LASSO	1.590 (4.3)	3.519 (3.1)	1.204 (0.0)*	0.791 (4.0)
full 30, random forest	1.479 (30.0)	3.443 (30.0)	1.086 (30.0)	0.831 (30.0)
pairwise-screened, random forest	1.474 (7.5)	3.429 (7.8)	1.079 (8.0)	0.847 (6.0)
ten counts m_{ij} , random forest	1.458 (10.0)	3.499 (10.0)	1.150 (10.0)	0.841 (10.0)
ten counts m_{ij} , LASSO	1.621 (8.8)	3.498 (7.0)	1.135 (8.6)	0.800 (9.0)
RDKit 2D descriptors, random forest	0.674 (159.7)	1.267 (148.9)	0.714 (160.2)	0.935 (172.5)
Comparison	difference [corrected 90% CI]			
pairwise vs full (LASSO)	+0.045 [+0.023, +0.066] eq.	−0.001 [−0.072, +0.071] eq.	+0.009 [+0.004, +0.013] eq.	−0.018 [−0.036, +0.001]
combined vs full (LASSO)	+0.055 [+0.031, +0.079]	+0.004 [−0.078, +0.085] eq.	+0.073 [+0.060, +0.086]	−0.017 [−0.035, +0.001]
pairwise vs full (RF)	−0.006 [−0.032, +0.021] eq.	−0.014 [−0.099, +0.071] eq.	−0.007 [−0.015, +0.000] eq.	+0.016 [+0.004, +0.028]
m_{ij} vs full 30 (RF)	−0.021 [−0.088, +0.046]	+0.056 [−0.187, +0.299]	+0.064 [+0.039, +0.089]	+0.010 [−0.005, +0.025]
RDKit 2D vs full 30 (RF)	−0.805 [−0.892, −0.718]	−2.176 [−2.441, −1.911]	−0.372 [−0.390, −0.354]	+0.105 [+0.084, +0.126]

Four results stand out. (i) The chemistry-aware reference is far ahead of every topology-only model — by 0.81 RMSE on ESOL, 2.18 on FreeSolv, 0.37 on Lipophilicity and 0.10 AUC on BBBP — and on ESOL and FreeSolv it exceeds, out of sample, the in-sample graph oracle of the topology-only representation. (ii) In a random forest the ten counts m_{ij} alone are not distinguishable from the full 30-index baseline on ESOL, FreeSolv and BBBP (intervals include zero, although on FreeSolv the interval is wide); only on Lipophilicity do the distance and spectral indices add measurably (+0.064 RMSE for the counts alone). (iii) Restricting the baseline to its pairwise-screened subset, and the combined screen, are discussed in Supplementary Section S4.

9 Discussion

Practical guidance.

- *Proposing a new degree-based index:* report its weight vector $(f(i, j))_{(i, j) \in P_4}$ and its coefficients on a classical basis (**certify**); do not report pairwise correlations as evidence of novelty; justify the index by parsimony, interpretation or mathematics, and compare its single-index R^2 with ordinary least squares on the ten counts.
- *Modelling with topological descriptors:* use the ten counts m_{ij} directly instead of many BID indices; expect topology-only descriptors to add little to a chemistry-aware descriptor set (≤ 0.024 RMSE here).
- *When the result does not apply:* indices outside the BID class (Remark 4.4), datasets with atoms of degree above four (Section 4.1), and non-linear models, for which only the empirical screen of Section 7 speaks.

A concrete benchmark for BID indices. Because every BID index lies in $\text{col}[\mathbf{1}, M]$, its in-sample R^2 in a single-index linear model can never exceed the R^2 of ordinary least squares on the ten counts m_{ij} . On our data that bound is 0.42 (ESOL), 0.19 (FreeSolv), 0.12 (Lipophilicity) and 0.26 (BBBP, as a linear probability model), while the best of the 18 baseline indices, the 15 published instances and the 100 Loyola points reaches 0.36, 0.13, 0.05 and 0.16 (**results/bid_r2_bound.csv**). A proposer of a new BID index can therefore report, on any

topological; their absence can only weaken the reference, which nevertheless leads by 0.37–2.18 RMSE.

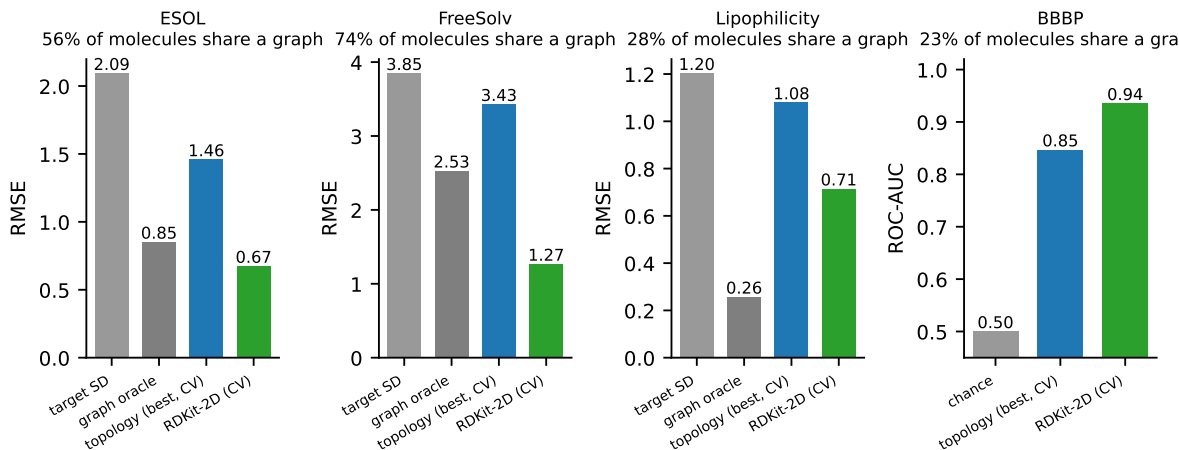


Figure 2: The information ceiling. Per dataset: the target standard deviation, the in-sample graph oracle (mean target of the isomorphism class; omitted for BBBP, where it is trivially near one), the best topology-only model of Table 7 (cross-validated), and the RDKit 2D random forest (cross-validated). On ESOL and FreeSolv the chemistry-aware model beats, out of sample, what any topology-only model could reach in sample.

dataset, how close the index comes to this bound; a modeller who wants the degree information should simply use the ten counts.

What a new degree-based index must show. On molecular graphs (maximum degree four) a new BID index cannot be linearly independent of a baseline that includes ten generic classical degree-based indices; this is a theorem, not an empirical finding. Its justification must therefore come from elsewhere: parsimony and interpretability (one number that summarises the ten counts for a given property), mathematical properties, or behaviour under models that are not linear in the descriptors. We suggest that a proposal of a new BID index report its coefficient vector $(f(i, j))_{(i, j) \in P_4}$, which places it exactly in the common ten-dimensional space and makes coincidences between differently named indices immediately visible.

What the target-aware screen adds. For indices outside the BID class the question is genuinely empirical, and the answer depends on the baseline and on uncertainty. Against a chemistry-aware baseline none passes (Table 6); against the topological baseline of the point-estimate “passers” in an earlier version of this analysis, none survives on ESOL or FreeSolv once intervals are reported, and most linear signals do not survive non-linear residualization. A pairwise threshold, which ignores both the multivariate baseline and the target, is not a meaningful novelty criterion.

What topology cannot do. Section 8 shows that a large share of molecules in these benchmarks are indistinguishable to every hydrogen-suppressed, unlabelled-graph descriptor. This is not a defect of any particular index; it bounds the whole class and explains the weak predictive performance of topological indices on targets such as FreeSolv.

10 Limitations

- The certificate concerns linear redundancy; it does not rank BID indices by usefulness, and it is silent on non-linear models except through Section 7.

- The target-aware analyses use four MoleculeNet benchmarks. The certificate itself was checked on thirteen; datasets with hypervalent or metal centres (HIV) leave $\mathcal{G}_4$ and require either a larger baseline or the counts m_{ij} themselves, which the data-level rank check of Proposition 4.2 detects.
- The graph oracle is an in-sample bound for a given dataset; the RDKit reference omits 28 descriptors for software reasons (footnote in Section 8).
- The effect-size margin $\tau = 0.10$ is a convention; we report intervals so that readers can apply their own.
- The non-linear check relies on random-forest residualization, whose null distribution for exactly redundant columns has a 99th percentile of 0.08–0.14 depending on the dataset; effects below that width are not resolved, and with 500 draws per dataset the smallest attainable p -value is 0.002, so a single pair can reach at best $q \approx 0.08$ after correction over 80 pairs.
- BBBP partial correlations use a linear probability model.
- The candidate set of Section 7 is illustrative, not exhaustive.

11 Conclusion

For bond-incident-degree indices on molecular graphs, redundancy is not a matter of thresholds: a rank condition on the baseline, met by any ten generic classical degree-based indices, makes every such index an exact linear combination of the baseline, with zero partial correlation with every target. We verified this on 13 MoleculeNet datasets (318 648 molecules; the one exception contains organometallic centres and is caught by the data-level check) and on fourteen published indices (15 instances), including eight recent Sombor-type variants, and showed that statistical diagnostics computed without this knowledge can produce spurious signal. For indices outside the class, a target-aware screen with intervals, a calibrated non-linear check and a chemistry-aware baseline is informative, and on these benchmarks it finds two suggestive signals against a topological baseline (each exceeding 500 exactly redundant null draws, neither reaching $q < 0.05$ over 80 pairs) and none against a chemistry-aware one. Both results sit under an information ceiling set by graph collisions, which limits every topology-only descriptor.

Data and code availability

Code, curated-data scripts, results and figures are available at <https://github.com/Singati2/topological-index-orthogonality>; the snapshot reported here is git tag v2.1-paper2. Datasets are downloaded from the MoleculeNet distribution by `src/load_data.py`. The analyses of this paper are produced by the following scripts in `scripts/`: 70 (certificate, census, identities), 02, 16 and 17 (screens), 73 (non-linear check), 74 (de-duplication), 72 (curation, collisions, benchmark), 76 (dataset verification), 77 (exact isomorphism check, R^2 bound), 78 (certificate on thirteen datasets), 79 (screen against the RDKit baseline), 80 (null distribution of ρ_{nl}), 81 (figures) and 75 (tables and the numerical macros used in the text); the one-function certificate is `src/certify.py`. The four datasets are included in `data/`, verified byte-identical to the MoleculeNet distribution. All analyses used Python 3.11, RDKit 2022.09, scikit-learn 1.8 and NumPy 2.4 with the seeds recorded in each script (20261005 unless stated).

Acknowledgements

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References

- [1] H. Wiener, *Structural determination of paraffin boiling points*, J. Am. Chem. Soc. **69**(1) (1947), 17–20. <https://doi.org/10.1021/ja01193a005>.
- [2] I. Gutman and N. Trinajstić, *Graph theory and molecular orbitals. Total π -electron energy of alternant hydrocarbons*, Chem. Phys. Lett. **17**(4) (1972), 535–538. [https://doi.org/10.1016/0009-2614\(72\)85099-1](https://doi.org/10.1016/0009-2614(72)85099-1).
- [3] M. Randić, *Characterization of molecular branching*, J. Am. Chem. Soc. **97**(23) (1975), 6609–6615. <https://doi.org/10.1021/ja00856a001>.
- [4] I. Gutman, *Degree-based topological indices*, Croat. Chem. Acta **86**(4) (2013), 351–361. <https://doi.org/10.5562/cca2294>.
- [5] I. Gutman, *Geometric approach to degree-based topological indices: Sombor indices*, MATCH Commun. Math. Comput. Chem. **86**(1) (2021), 11–16.
- [6] V. R. Kulli and I. Gutman, *Computation of Sombor indices of certain networks*, SSRG Int. J. Appl. Chem. **8**(1) (2021), 1–5. <https://doi.org/10.14445/23939133/IJAC-V8I1P101>.
- [7] I. Gutman, B. Furtula, and M. S. Oz, *Geometric approach to vertex-degree-based topological indices – Elliptic Sombor index, theory and application*, Int. J. Quantum Chem. **124**(2) (2024), e27346. <https://doi.org/10.1002/qua.27346>.
- [8] I. Gutman, *Relating Sombor and Euler indices*, Vojnoteh. Glas. **72**(1) (2024), 1–12. <https://doi.org/10.5937/vojtehg72-48818>.
- [9] F. Movahedi, I. Gutman, I. Redžepović, and B. Furtula, *Diminished Sombor index*, MATCH Commun. Math. Comput. Chem. **95**(1) (2026), 141–162. <https://doi.org/10.46793/match95-1.14125>.
- [10] J. Barman and S. Das, *Geometric approach to degree-based topological index: Hyperbolic Sombor index*, MATCH Commun. Math. Comput. Chem. **95**(1) (2026), 63–94. <https://doi.org/10.46793/match95-1.03425>.
- [11] T. Réti, T. Došlić, and A. Ali, *On the Sombor index of graphs*, Contrib. Math. **3** (2021), 11–18. <https://doi.org/10.47443/cm.2021.0006>.
- [12] V. R. Kulli, N. Harish, B. Chaluvaram, and I. Gutman, *Mathematical properties of KG Sombor index*, Bull. Int. Math. Virtual Inst. **12**(2) (2022), 379–386. <https://doi.org/10.7251/BIMVI2202379K>.
- [13] B. Furtula, I. Gutman, and M. Dehmer, *On structure-sensitivity of degree-based topological indices*, Appl. Math. Comput. **219**(17) (2013), 8973–8978. <https://doi.org/10.1016/j.amc.2013.03.072>.
- [14] V. Kumar and S. Das, *On structure sensitivity and chemical applicability of some novel degree-based topological indices*, MATCH Commun. Math. Comput. Chem. **92**(1) (2024), 165–203. <https://doi.org/10.46793/match.92-1.165K>.
- [15] M. Dehmer, F. Emmert-Streib, and S. Tripathi, *Large-scale evaluation of molecular descriptors by means of clustering*, PLoS ONE **8**(12) (2013), e83956. <https://doi.org/10.1371/journal.pone.0083956>.
- [16] Y. Ma, M. Dehmer, U.-M. Künzi, S. Tripathi, M. Ghorbani, J. Tao, and F. Emmert-Streib, *The usefulness of topological indices*, Inform. Sci. **606** (2022), 143–151. <https://doi.org/10.1016/j.ins.2022.05.038>.
- [17] J. Rada and R. Cruz, *Vertex-degree-based topological indices over graphs*, MATCH Commun. Math. Comput. Chem. **72**(3) (2014), 603–616.

- [18] R. Cruz, C. Espinal, and J. Rada, *A matrix approach to vertex-degree-based topological indices*, Mathematics **12**(13) (2024), Article 2043. <https://doi.org/10.3390/math12132043>.
- [19] J. Rada, *A finite-dimensional linear framework for vertex-degree-based topological indices*, MATCH Commun. Math. Comput. Chem. **96**(3) (2026), 841–863. <https://doi.org/10.46793/match.96-3.32925>.
- [20] B. Hollas, *An analysis of the redundancy of graph invariants used in chemoinformatics*, Discrete Appl. Math. **154**(17) (2006), 2484–2498. <https://doi.org/10.1016/j.dam.2006.04.002>.
- [21] B. Hollas, *The covariance of topological indices that depend on the degree of a vertex*, MATCH Commun. Math. Comput. Chem. **54**(1) (2005), 177–187.
- [22] B. Lučić and N. Trinajstić, *New developments in QSPR/QSAR modeling based on topological indices*, SAR QSAR Environ. Res. **7**(1–4) (1997), 45–62. <https://doi.org/10.1080/10629369708039124>.
- [23] Y. Du, Y. Liang, B. Li, and C. Xu, *Orthogonalization of block variables by subspace-projection for quantitative structure property relationship (QSPR) research*, J. Chem. Inf. Comput. Sci. **42**(5) (2002), 993–1003. <https://doi.org/10.1021/ci020283+>.
- [24] I. Redžepović, *Chemical applicability of Sombor indices*, J. Serb. Chem. Soc. **86**(5) (2021), 445–457. <https://doi.org/10.2298/JSC201215006R>.
- [25] D. Vukičević and M. Gašperov, *Bond additive modeling 1. Adriatic indices*, Croat. Chem. Acta **83**(3) (2010), 243–260. <https://hrcak.srce.hr/clanak/93215>.
- [26] D. Vukičević and J. Đurđević, *Bond additive modeling 10. Upper and lower bounds of bond incident degree indices of catacondensed fluoranthenes*, Chem. Phys. Lett. **515**(1–3) (2011), 186–189. <https://doi.org/10.1016/j.cplett.2011.08.095>.
- [27] E. Deutsch and S. Klavžar, *M-polynomial and degree-based topological indices*, Iranian J. Math. Chem. **6**(2) (2015), 93–102. <https://doi.org/10.22052/ijmc.2015.10106>.
- [28] I. Gutman, J. Buragohain, and G. G. Nyauli, *Concealed relations between vertex-degree-based topological indices*, J. Math. Chem. **64**(6) (2026), Article 39. <https://doi.org/10.1007/s10910-026-01799-0>.
- [29] V. Dusollier, S. Bonte, G. Devillez, A. Hertz, H. Mélot, and D. Schindl, *Complete polyhedral description of chemical graphs of maximum degree at most 3*, arXiv:2506.19768 (2025). <https://arxiv.org/abs/2506.19768>.
- [30] S. Bonte, G. Devillez, V. Dusollier, A. Hertz, H. Mélot, and D. Schindl, *ChemicHull: an online tool for determining extremal chemical graphs of maximum degree at most 3 for any degree-based topological indices*, Discrete Math. Chem. (2026). <https://doi.org/10.26493/2820-6657.9.e6f>.
- [31] S. Bonte, G. Devillez, V. Dusollier, A. Hertz, and H. Mélot, *Extremal chemical graphs of maximum degree at most 3 for 33 degree-based topological indices*, MATCH Commun. Math. Comput. Chem. **95**(1) (2025), 193–231. <https://doi.org/10.46793/match.95-1.00125>.
- [32] M. G. Schuh, A. Daniluk, and S. A. Sieber, *Auditing widely used biomolecular benchmarks reveals systematic data inconsistencies*, Chem. Sci. **17**(36) (2026), 17455–17467. <https://doi.org/10.1039/d6sc01799a>.
- [33] E. V. Konstantinova, *The discrimination ability of some topological and information distance indices for graphs of unbranched hexagonal systems*, J. Chem. Inf. Comput. Sci. **36**(1) (1996), 54–57. <https://doi.org/10.1021/ci9502461>.
- [34] L. Pogliani, *Core electrons and hydrogen atoms in chemical graph theory*, J. Math. Chem. **43**(3) (2008), 1233–1255. <https://doi.org/10.1007/s10910-007-9253-7>.
- [35] M. Dehmer, M. Grabner, and K. Varmuza, *Information indices with high discriminative power for graphs*, PLoS ONE **7**(2) (2012), e31214. <https://doi.org/10.1371/journal.pone.0031214>.

- [36] J. Adamczyk and W. Czech, *Molecular Topological Profile (MOLTOP) – simple and strong baseline for molecular graph classification*, in: ECAI 2024, Frontiers in Artificial Intelligence and Applications **392**, IOS Press, 2024, pp. 1575–1582. <https://doi.org/10.3233/FAIA240663>. (arXiv:2407.12136)
- [37] M. Praski, J. Adamczyk, and W. Czech, *Benchmarking pretrained molecular embedding models for molecular representation learning*, arXiv:2508.06199 (2025). <https://arxiv.org/abs/2508.06199>.
- [38] R. Tibshirani, *Regression shrinkage and selection via the lasso*, J. R. Stat. Soc. B **58**(1) (1996), 267–288. <https://doi.org/10.1111/j.2517-6161.1996.tb02080.x>.
- [39] H. Peng, F. Long, and C. Ding, *Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy*, IEEE Trans. Pattern Anal. Mach. Intell. **27**(8) (2005), 1226–1238. <https://doi.org/10.1109/TPAMI.2005.159>.
- [40] M. B. Kursu and W. R. Rudnicki, *Feature selection with the Boruta package*, J. Stat. Softw. **36**(11) (2010), 1–13. <https://doi.org/10.18637/jss.v036.i11>.
- [41] Z. Wu, B. Ramsundar, E. N. Feinberg, J. Gomes, C. Geniesse, A. S. Pappu, K. Leswing, and V. Pande, *MoleculeNet: a benchmark for molecular machine learning*, Chem. Sci. **9**(2) (2018), 513–530. <https://doi.org/10.1039/c7sc02664a>.
- [42] J. S. Delaney, *ESOL: Estimating aqueous solubility directly from molecular structure*, J. Chem. Inf. Comput. Sci. **44**(3) (2004), 1000–1005. <https://doi.org/10.1021/ci034243x>.
- [43] D. L. Mobley and J. P. Guthrie, *FreeSolv: a database of experimental and calculated hydration free energies, with input files*, J. Comput. Aided Mol. Des. **28**(7) (2014), 711–720. <https://doi.org/10.1007/s10822-014-9747-x>.
- [44] AstraZeneca, *Experimental in vitro DMPK and physicochemical data on a set of publicly disclosed compounds*, ChEMBL document ChEMBL3301361 (2015), <https://doi.org/10.6019/CHEMBL3301361>; distributed via MoleculeNet [41]. Curated set of 4 200 compound–log $D_{7.4}$ pairs.
- [45] I. F. Martins, A. L. Teixeira, L. Pinheiro, and A. O. Falcao, *A Bayesian approach to in silico blood-brain barrier penetration modeling*, J. Chem. Inf. Model. **52**(6) (2012), 1686–1697. <https://doi.org/10.1021/ci300124c>.
- [46] G. Landrum et al., *RDKit: Open-source cheminformatics*, <https://www.rdkit.org>.
- [47] D. Vukićević and B. Furtula, *Topological index based on the ratios of geometrical and arithmetical means of end-vertex degrees of edges*, J. Math. Chem. **46**(4) (2009), 1369–1376. <https://doi.org/10.1007/s10910-009-9520-x>.
- [48] M. O. Albertson, *The irregularity of a graph*, Ars Combin. **46** (1997), 219–225.
- [49] V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins, *Double/debiased machine learning for treatment and structural parameters*, Econom. J. **21**(1) (2018), C1–C68. <https://doi.org/10.1111/ectj.12097>.
- [50] R. D. Shah and J. Peters, *The hardness of conditional independence testing and the generalised covariance measure*, Ann. Statist. **48**(3) (2020), 1514–1538. <https://doi.org/10.1214/19-AOS1857>.
- [51] N. Shervashidze, P. Schweitzer, E. J. van Leeuwen, K. Mehlhorn, and K. M. Borgwardt, *Weisfeiler–Lehman graph kernels*, J. Mach. Learn. Res. **12** (2011), 2539–2561. <https://www.jmlr.org/papers/v12/shervashidze11a.html>.
- [52] C. Nadeau and Y. Bengio, *Inference for the generalization error*, Mach. Learn. **52**(3) (2003), 239–281. <https://doi.org/10.1023/A:1024068626366>.
- [53] D. J. Schuirmann, *A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability*, J. Pharmacokinet. Biopharm. **15**(6) (1987), 657–680. <https://doi.org/10.1007/BF01068419>.

- [54] V. R. Kulli, *Nirmala index*, Int. J. Math. Trends Technol. **67**(3) (2021), 8–12. <https://doi.org/10.14445/22315373/IJMTT-V67I3P502>.
- [55] V. R. Kulli, V. Loksha, and K. Nirupadi, *Computation of inverse Nirmala indices of certain nanostructures*, Int. J. Math. Combin. **2** (2021), 33–40.
- [56] V. R. Kulli, *The Gourava indices and coindices of graphs*, Ann. Pure Appl. Math. **14**(1) (2017), 33–38. <https://doi.org/10.22457/apam.v14n1a4>.
- [57] G. H. Shirdel, H. Rezapour, and A. M. Sayadi, *The hyper-Zagreb index of graph operations*, Iranian J. Math. Chem. **4**(2) (2013), 213–220. <https://doi.org/10.22052/ijmc.2013.5294>.